

Hibernate Many-to-Many Association

- **Definition:** A many-to-many association refers to a relationship where multiple instances of one entity are associated with multiple instances of another entity.

Example in Code:

```
package com.telusko.model;

import java.util.Set;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.ManyToMany;

@Entity
public class Students
{
    @Id
    @Column(name="students_id")
    private Integer id;

    private String name;

    private String city;

    @ManyToMany(cascade=CascadeType.ALL)
    private Set<Courses> courses;

    public Set<Courses> getCourses() {
        return courses;
    }
}
```

```
}
```

```
public void setCourses(Set<Courses> courses) {
```

```
    this.courses = courses;
```

```
}
```

```
public Students()
```

```
{
```

```
    System.out.println("Zero Param Constructor of Students ");
```

```
}
```

```
public Integer getId() {
```

```
    return id;
```

```
}
```

```
public void setId(Integer id) {
```

```
    this.id = id;
```

```
}
```

```
public String getName() {
```

```
    return name;
```

```
}
```

```
public void setName(String name) {
```

```
    this.name = name;
```

```
}
```

```
public String getCity() {
```

```
    return city;
```

```
}
```

```
public void setCity(String city) {
```

```
    this.city = city;
```

```
}
```

```
        @Override
        public String toString() {
            return "Students [id=" + id + ", name=" + name + ", city=" +
city + ", courses=" + courses + "]";
        }
    }
}
```

```
package com.telusko.model;

import jakarta.persistence.CascadeType;
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.OneToOne;

@Entity
public class Courses
{
    @Id
    @Column(name="courses_id")
    private Integer id;

    private String courseName;

    private Integer coursePrice;

    public Courses()
    {
        System.out.println("Zero Param Constructor of Courses");
    }
}
```

```
public Integer getCoursePrice() {  
    return coursePrice;  
}  
  
public void setCoursePrice(Integer coursePrice) {  
    this.coursePrice = coursePrice;  
}  
  
public Integer () {  
    return id;  
}  
  
public void setId(Integer id) {  
    this.id = id;  
}  
  
public String getCourseName() {  
    return courseName;  
}  
  
public void setCourseName(String courseName) {  
    this.courseName = courseName;  
}  
  
@Override  
public String toString() {  
    return "Courses [id=" + id + ", courseName=" + courseName + ",  
coursePrice=" + coursePrice + "];"  
}  
  
}
```

```
package com.telusko.app;

import java.util.HashSet;
import java.util.Set;

import org.hibernate.HibernateException;
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.Transaction;
import org.hibernate.cfg.Configuration;

import com.telusko.model.Courses;
import com.telusko.model.Students;

public class LaunchStandardApp
{

    public static void main(String[] args)
    {
        Configuration config = null;
        SessionFactory sessionFactory = null;
        Session session = null;
        Transaction transaction = null;
        boolean flag=false;

        config=new Configuration();

        config.configure();

        sessionFactory=config.buildSessionFactory();

        session=sessionFactory.openSession();

        Courses c1=new Courses();
```

```
c1.setId(1);  
c1.setCourseName("Java");  
c1.setCoursePrice(1999);
```

```
Courses c2=new Courses();  
c2.setId(2);  
c2.setCourseName("Hibernate");  
c2.setCoursePrice(499);
```

```
Courses c3=new Courses();  
c3.setId(3);  
c3.setCourseName("SpringBoot");  
c3.setCoursePrice(999);
```

```
Set<Courses> set1=new HashSet<>();  
set1.add(c1);  
set1.add(c2);  
set1.add(c3);
```

```
Set<Courses> set2=new HashSet<>();  
set2.add(c1);  
set2.add(c2);
```

```
Students s1 = new Students();  
s1.setId(1);  
s1.setName("Ramesh");  
s1.setCity("Chennai");  
s1.setCourses(set1);
```

```
Students s2 = new Students();  
s2.setId(2);  
s2.setName("Harsh");  
s2.setCity("Jaipur");
```

```

s2.setCourses(set2);

Students s3 = new Students();
s3.setId(3);
s3.setName("Shushil");
s3.setCity("Mumbai");
s3.setCourses(set1);

//
//      Students stu1 = session.get(Students.class, 1);
//      System.out.println(stu1);
//      System.out.println("*****");
//
//      Students stu2 = session.get(Students.class, 2);
//      System.out.println(stu2);
//
//      System.out.println("*****");
//
//      Students stu3 = session.get(Students.class, 3);
//      System.out.println(stu3);
//

try
{
    transaction=session.beginTransaction();
    session.persist(s1);
    session.persist(s2);
    session.persist(s3);
    flag=true;
}
catch(HibernateException e)
{
    e.printStackTrace();
}

```

```
    }  
    catch(Exception e)  
    {  
        e.printStackTrace();  
    }  
  
    finally  
    {  
        if(flag==true)  
        {  
            transaction.commit();  
        }  
        else  
        {  
            transaction.rollback();  
        }  
  
        session.close();  
        sessionFactory.close();  
    }  
}  
  
}
```


Output:

```
mysql> create database telusko_db;  
Query OK, 1 row affected (0.02 sec)
```

```
mysql> use telusko_db;  
Database changed
```

```
mysql> show tables;
```

```
+-----+  
| Tables_in_telusko_db |  
+-----+  
| courses               |  
| students              |  
| students_courses      |  
+-----+  
3 rows in set (0.01 sec)
```

```
mysql> select * from courses;
+-----+-----+-----+
| courses_id | courseName | coursePrice |
+-----+-----+-----+
|          1 | Java      |          1999 |
|          2 | Hibernate |           499 |
|          3 | SpringBoot |           999 |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from students;
+-----+-----+-----+
| students_id | city      | name      |
+-----+-----+-----+
|           1 | Chennai  | Ramesh    |
|           2 | Jaipur   | Harsh     |
|           3 | Mumbai   | Shushil   |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from students_courses;
+-----+-----+-----+
| Students_students_id | courses_courses_id |
+-----+-----+-----+
|                     1 |                   1 |
|                     2 |                   1 |
|                     3 |                   1 |
|                     1 |                   2 |
|                     2 |                   2 |
|                     3 |                   2 |
|                     1 |                   3 |
|                     3 |                   3 |
+-----+-----+-----+
```

- **Explanation:**

- The @ManyToMany annotation maps a many-to-many relationship between Student and Course.
- A Set<Course> is used to represent the list of courses a student is enrolled in.

- CascadeType ALL ensures that all operations performed on the Student entity cascade to the associated Course entities.

Use Cases:

- Multiple students can enroll in multiple courses, and each course can have multiple students.